

# Quaxtrip Manual

Quaxtrip is a set of Max patches that makes low-latency uncompressed audio and messaging interconnections over the internet, intended for musicians wishing to play together remotely.

## Background

Quaxtrip runs Miller Puckette's Quacktrip Pure Data patch within Cycling '74's Max. Quacktrip, in turn, is an implementation, in Pure Data, of Chris Chafe's Jacktrip network protocol, based on jacktrip.pd by Roman Haefeli and Johannes Schuett. It establishes a low-latency, point-to-point connection between two sites, with no audio compression.

## Installation

1. If you don't already have Max 8, [download](#) and install it. Note that you do not need to purchase Max to use Quaxtrip, it will run in the unregistered version.
2. [Download](#) and install Pure Data (version 0.55-0 or later — listed on the download page under **Current Stable Release**).
3. If Max is running, quit Max.
4. [Download](#) PdMax (version 0.54 or greater — listed on the download page under **Experimental: pd~ ported to Max/MSP 6,7, and 8 by Puckette & Apel**), unzip, and place in the ~/Documents/Max 8/library directory. PdMax contains the Max object (pd~) that allows you to run Pd patches in Max. *Note: users of Apple Silicon Macs (M1 processor or later) will need to run Max in Rosetta mode when using PdMax 0.54 (get info on app and check Open Using Rosetta). If you prefer not to run in Rosetta mode, there is a beta version of PdMax on the download page you may try (0.55test1).*
5. [Download](#) the latest release of Quaxtrip, unzip, and place in the ~/Documents/Max 8/library directory.

## Quick Start

There are two versions of Quaxtrip: **Quaxtrip Duet.maxpat** and **Quaxtrip Quintet.maxpat**. The only difference between these two, as the names imply, is that **Duet** allows only one remote partner while **Quintet** allows up to four. If you are playing with only one partner, **Duet** frees up some screen space and uses somewhat less CPU.

*Note: It is **highly** recommended that you are connected to the internet via Ethernet rather than WiFi. Even a "fast" WiFi connection is likely to be unreliable for realtime work.*

How to play with a remote partner:

1. Both you and your remote partner launch **Quaxtrip Duet.maxpat**.
2. If this is the first time you've launched Quaxtrip, you will be presented with a dialog box instructing you to locate and select the Pd app (binary) on your computer. You will only need to do this the first time you run the patch.
  - a. macOS: The Pd app is typically in the /Applications directory.
  - b. Windows: The Pd app is typically in the /Program Files/Pd/bin directory.
3. In the **Local Input** channel (top left corner of the patch):
  - a. Set the **Input Channels** to reflect the channel(s) you are receiving incoming audio from your audio interface. If you are only using 1 channel, check the **Mono** checkbox.
  - b. Set volume as appropriate.
4. Adjust the **Output** channel volume to make it audible. You should now hear yourself. If using live mics, watch out for feedback.
5. In the **Remote 1** channel:
  - a. Quaxtrip uses the **Server** foo.ucsd.edu by default. This public "conniption" server helps make the connection but once connected the audio runs directly between the two computers.
  - b. Set the **Call Name** to something likely to be unique (e.g. jack-and-jill), no spaces allowed. The **Call Name** is the same for you and your remote partner.
6. Set the **On/Off** toggle to on.

Once you and your partner have completed these steps, the connected checkbox should be checked and you should now be able to hear each other! If you're experiencing dropouts, see below for how to fine-tune the settings.

If you want to play with more than one partner, use **Quaxtrip Quintet.maxpat** instead and repeat steps 5 & 6 for each partner, making sure to select a unique **Call Name** for each set of partners.

## Advanced Settings

- The **Local Input** has two mute checkboxes. **Monitor Mute** silences your input locally (i.e. you won't hear it out of the **Monitor Output**). **Remote Mute** prevents your signal from being sent to remote partners, but you can still hear it locally. **Monitor Mute** is also available on the **Remote** channels.
- If your signal is mono, set **Channels** to 1 to save bandwidth. Note that this need not be symmetrical. That is, you could send 1 channel while your partner sends you 2.
- The **Input Delay** can be set to absorb network time variations (in units of 64 samples). Maximum **Input Delay** is 100. The **Fill** counter shows how far ahead of the incoming stream you are. If this number drops below 2 there will be a dropout in the audio output. Increasing the **Input Delay** should decrease dropouts at the expense of latency.
- **Block Size** sets the size of the chunks of the audio you send. Increasing this number may decrease dropouts at the expense of latency.

- Check **2X** to send each outgoing packet twice (this doubles the data rate but might give fewer dropped packets).
- Be sure your sample rate (set in **Options -> Audio Status** in Max) matches your partner's - if they don't you'll get distorted sound.
- If the **Calling Server** button is flashing you're either waiting for a response from the server, or from the partner. Once both of these arrive the **Connected** toggle turns on. At this point a call is in progress and the "packets in" counter should be changing. Press the **Reset** button to set **Packets In** and **Dropouts** counts back to zero.
- Quaxtrip automatically sets Max's **Signal Vector Size** to 64 and engages **Scheduler in Overdrive**. Changes to these settings may result in distortion and crashes.
- To test latency:
  - a. Both you and a remote partner click **Latency Tester** in the appropriate **Remote** channel.
  - b. One partner selects **Measure** in the **Latency** window pop-up.
  - c. The other partner selects **Loop-back** in the **Latency** window pop-up.
  - d. The latency reading will show up in the window for both partners.
- Click the **Save Settings** button to save the current server, volume, etc. settings to the location of your choice on your hard drive. Click the **Load Settings** button to load a previous saved setup.

## Messages

Quaxtrip includes a rudimentary messaging system to allow the sending of text messages to the other participants. Click the **Messages** button and type a message in the pop-up window to send a text message to all connected peers. Text messages received appear in the Max Console.

## Mixer

Quaxtrip contains a mixer that allows you to mix up to 8 stereo or mono signals from your computer's audio interface. To launch the **Mixer**, select **Quaxtrip Mixer** in the **Input Channels** pull-down menu in the **Local Input** section. This will open the **Mixer** window and set the **Local Input** to receive the mix. The **Mixer** has 8 channels. Choose the input channel(s) at the top of each channel. Each channel can be individually set as stereo or mono.

## Recorder

The Quaxtrip **Recorder** allows you to make synchronized recordings of each peer's local signal. This allows all peers to make a local recording free of any glitches that might arise over the internet.

To set up the **Recorder**, each participant needs to select their settings:

- Click the **Recorder** button in the main **Quaxtrip Duet** or **Quaxtrip Quintet** patch.

- Set the desired **File Type**, **Sample Rate**, and **Sample Type** from the pull-down menus in the **Quaxtrip Recorder** window.
- If you are using the **Local Bus** in the **Local Input** channel, you will see a checkbox labeled **Pre-local Bus**. Toggle this to on if you wish to record the dry signal only (i.e. the input signal before it is sent to the bus). With this set to off you will record the **Local Input** signal after it is returned from the **Local Bus** (i.e as heard in the **Monitor Output**).

When all peers are ready, toggle on the **Start/Stop Recording** button. This will start the local recording as well as all currently connected **Remote** partners.

Recordings are saved in the **recordings** folder of the Quaxtrip folder, labeled with the date and time of recording.

## Using the Connption Server

By default quacktrip uses a ‘connption’ server at foo.ucsd.edu, port 3840 (hexadecimal “f00”) to connect calls. This should work for most people in most situations. You can, however, enter an alternative server name or IP address in the **Server** box if you wish. A connption server is built into Quaxtrip.

### Computers on the Same Local Network

If you want to make calls between computers behind the same router (i.e. your local network), you will need to run a connption server behind that same router. Calls between two patches on the same computer won’t work. On the computer you wish to be the server, click the **Connption Server** button to view your IP address. Set the **Server** of all computers to that listed as the **Local IP Address** in the **Connption Server** window. If you see more than one **Local IP Address** listed that means you have more than one local connection, probably both Ethernet and WiFi. Either turn off WiFi to avoid the ambiguity or check your computer’s network settings to find out which IP address is currently active.

### Over the Internet

If using your computer as a connption server with other computers over the internet, set the **Server** of all computers to that listed as the **Public IP Address** in the **Connption Server** window of the computer functioning as a server. Before this will work, you will need to [set up port forwarding](#) for port 3840 on the server computer. Note that port forwarding is only necessary when connecting computers over the internet, not for computers on the same local network, nor is it necessary if you’re using the default foo.ucsd.edu connption server.

# Integrating Quaxtrip into Your Own Patches

## Routing Audio to and from other Max Patches

Quaxtrip includes several send~ and receive~ objects intended to allow you to send audio to and/or from your own Max patches. While Quaxtrip is open, open **Quaxtrip Routing.maxpat** in Quaxtrip's lib directory to see a demonstration of the various ins and outs available.

### Sending Audio to the Local Channel

Using **qxt\_local\_input\_L** and **qxt\_local\_input\_R** send~ objects in your patch you can send audio from your patch to the **Local** channel, bypassing the **Local** audio input.

To set Quaxtrip to receive this audio:

1. Select send **qxt\_local\_input** in the **Input Starting Channel** pull-down of the **Local** channel.
2. Include **qxt\_local\_input\_L** and **qxt\_local\_input\_R** send~ objects in your patch to send the audio.

### Using the Local Bus

Using receive~ and send~ objects in your patch you can route audio from the **Local** channel to your own patch and back again. These sends are pre-fader.

To use this effects loop:

1. Select the proper channel for your audio interface in the **Input Starting Channel** pull-down in the **Local** channel.
2. Check the **Local Bus** checkbox in the **Local** channel.
3. Include **qxt\_local\_bus\_send\_L** and **qxt\_local\_bus\_send\_R** receive~ objects in your patch to receive the audio.
4. Include **qxt\_local\_bus\_receive\_L** and **qxt\_local\_bus\_receive\_R** send~ objects in your patch to send the audio back to Quaxtrip.

### Channels - Monitor

These are send~ objects that route post-fader audio for the **Local Input** and for each **Remote** channel. All channels are post **Monitor Mute**.

To set your patch to receive audio from individual channels:

1. Add **0\_qxt\_monitor\_L** and **0\_qxt\_monitor\_R** receive~ objects in your patch to receive the Local channel.
2. Add **#\_qxt\_monitor\_L** and **#\_qxt\_monitor\_R** for each Remote channel (e.g. **1\_qxt\_monitor\_L** and **1\_qxt\_channel\_R** for Remote 1).

## Channels - Main

There are send~ objects that route post-fader audio for the **Local Input** and for each **Remote** channel. The **Local Input** send~ is post **Remote Mute**.

To set your patch to receive audio from individual channels:

3. Add **0\_qxt\_channel\_L** and **0\_qxt\_channel\_R** receive~ objects in your patch to receive the Local channel.
4. Add **#\_qxt\_channel\_L** and **#\_qxt\_channel\_R** for each Remote channel (e.g. **1\_qxt\_channel\_L** and **1\_qxt\_channel\_R** for Remote 1).

## Monitor Mix

The audio that is being sent out of the **Monitor Output** of Quaxtrip is available via send~ objects. These sends are post **Monitor Mute** and reflect the audio that Quaxtrip is sending to the dac~.

To set your patch to receive the **Monitor Output** audio:

1. Add **qxt\_monitor\_L** and **qxt\_monitor\_R** receive~ objects in your patch.

## Main Mix

A mix of all remote signals combined with the **Local Input** is available via send~ objects. The **Local Input** signal is post **Remote Mute**. This mix reflects what is heard by all the ensemble members.

To set your patch to receive the main mix audio:

1. Add **qxt\_mix\_L** and **qxt\_mix\_R** receive~ objects in your patch.

## Using the Messaging System

In addition to the audio that is sent between partners, there is also a system for sending messages. Messages can be symbols, numbers, or lists. While Quaxtrip is open, open **Quaxtrip Routing.maxpat** in Quaxtrip's lib directory and select the **Messages** tab to see a demonstration.

To send messages:

1. Add a **qxt\_send\_message** send object in your patch.
2. Send the message **n-message the message** where **n** is the **Remote** channel number where you wish to send the message and **the message** is the symbol, number, or list you wish to send.

To receive messages:

1. Add a **qxt\_receive\_message** receive object in your patch.
2. Messages received will be in the format **n the message** where **n** is the **Remote** channel number that sent the message and **the message** is the symbol, number, or list sent.

Quaxtrip Copyright 2020-2024 Damon Holzborn. Quaxtrip is open source software, which you can use and modify under the MIT License (LICENSE.txt in this distribution).

Also included in this distribution is Quacktrip. Quacktrip is Copyright 2020 Miller Puckette. Quacktrip is open source software. free to use and modify under the Standard Improved BSD License (quacktrip/lib/LICENSE.txt in this distribution).